# Hints: Number theory – 8

Homework

*The hints to each problem are cumulative. Read one, try again, and open the next only if you are still stuck.*

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## Problem 1

**Hint 1.** Use Vieta jumping (the Markov tree). From a solution $(x, y, z)$, the triple $(x, y, 3xy - z)$ is again a solution. Start from $(1, 1, 1)$ and always replace the smallest coordinate.

**Hint 2.** $(1,1,1) \to (1,1,2) \to (1,2,5) \to (1,5,13) \to (5,13,194) \to (13,194,7561) \to (194,7561,4400489)$. The last triple has all entries $> 50$.

## Problem 2

**Hint 1.** Look for $n$ with $n+1$ a perfect square: then $n+1 = x^2 + 0^2$ and $n+2 = x^2 + 1^2$, and only $n = x^2 - 1$ remains.

**Hint 2.** Make $x^2 - 1$ equal to $2y^2 = y^2 + y^2$. The Pell equation $x^2 - 2y^2 = 1$ has infinitely many solutions.

## Problem 3

**Hint 1.** The infinite continued fraction $x$ satisfies $x = 2207 - \frac{1}{x}$, so $x = \frac{2207 + \sqrt{2207^2 - 4}}{2}$ (the larger root, since $x > 1$).

**Hint 2.** Recognise 2207 as a Lucas number: with $\varphi = \frac{1+\sqrt{5}}{2}$, $\varphi^k + \varphi^{-k} = L_k$, and $L_{16} = 2207$. So $x = \varphi^{16}$, and the answer is $\varphi^2 = \frac{3+\sqrt{5}}{2}$.

## Problem 4

**Hint 1.** Solve the recurrence: $a_n = \frac{(2+\sqrt{3})^n + (2-\sqrt{3})^n}{2}$, i.e. $a_n = T_n(2)$ for the Chebyshev polynomial $T_n$.

**Hint 2.** For odd $k$, $a_m \mid a_{mk}$, because $\cos(k\theta)$ is a polynomial in $\cos\theta$ with no constant term when $k$ is odd (equivalently $x^k + y^k$ is divisible by $x + y$). Use $2015 = 5 \cdot 403$.

**Hint 3.** $a_5 = 362 = 2 \cdot 181$, so 181 divides $a_{2015}$. ($a_{13}$ and $a_{31}$ give other valid answers.)

## Problem 5

**Hint 1.** The coefficients satisfy $a_{n+2} = 2a_{n+1} + a_n$, $a_0 = 1$, $a_1 = 2$: they are the Pell numbers, $a_n = \frac{(1+\sqrt{2})^{n+1} - (1-\sqrt{2})^{n+1}}{2\sqrt{2}}$.

**Hint 2.** Prove $a_n^2 + a_{n+1}^2 = a_{2n+2}$, either from the closed form (using $(1+\sqrt{2})(1-\sqrt{2}) = -1$) or by induction via the matrix $\begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix}^n$. So $m = 2n + 2$.

## Problem 6

**Hint 1.** $a_{n+1}$ is the larger root of $t^2 - 7a_n t + a_n^2 + 9 = 0$. This relation is symmetric in $t$ and $a_n$, so $a_{n-1}$ is the other (smaller) root.

**Hint 2.** Vieta: $a_{n+1} + a_{n-1} = 7a_n$. With $a_0 = 1$, $a_1 = 5$, induction gives integrality (in fact $a_n = F_{4n+1}$).

**Hint 3.** From $a_{n+1}^2 + a_n^2 + 9 = 7a_n a_{n+1}$ you get $(a_n + a_{n+1})^2 = 9(a_n a_{n+1} - 1)$. Hence $a_n a_{n+1} - 1 = \left(\frac{a_n + a_{n+1}}{3}\right)^2$, and $3 \mid a_n + a_{n+1}$ follows from $9 \mid (a_n + a_{n+1})^2$.

## Problem 7

**Hint 1.** If a prime $p$ divides two coordinates, it divides the third. Then it divides all three after any Vieta jump, so the property would survive the descent down to $(1, 1, 1)$. Hence the coordinates are pairwise coprime.

**Hint 2.** $z \mid x^2 + y^2$ with $\gcd(x, y) = 1$. So $4 \nmid z$, since $x^2 + y^2 \equiv 1, 2 \pmod 4$, and $-1$ is a quadratic residue modulo every odd prime $p \mid z$, i.e. $p \equiv 1 \pmod 4$.

**Hint 3.** For $x^2 + y^2 + z^2 = kxyz$, take a solution with $x \leqslant y \leqslant z$ and minimal sum. The other root $z' = kxy - z$ must satisfy $z' \geqslant z$, hence $z^2 \leqslant zz' = x^2 + y^2 \leqslant 2y^2$. Plug in: $kxy^2 \leqslant kxyz = x^2 + y^2 + z^2 \leqslant 2(x^2 + y^2) \leqslant 4y^2$, so $kx \leqslant 4$. Here $kx = 4$ would force $x = y = z$ and $3x^2 = 4x^2$, so in fact $kx \leqslant 3$.

**Hint 4.** Cases: $k = 3, x = 1$ (Markov); $k = 1, x \leqslant 3$; $k = 2, x = 1$, which gives $(y - z)^2 = -1$, impossible. For $k = 1$, look modulo 3 at any solution: squares are $0, 1$, and checking how many of $x, y, z$ are divisible by 3 shows $3 \mid x, y, z$. Then $(x/3, y/3, z/3)$ is a Markov triple; conversely $(3x, 3y, 3z)$ always works.

## Problem 8

**Hint 1.** Multiplying by $\pm\varepsilon^k$ moves any solution $\beta > 0$ into $[\sqrt{|n|}, \varepsilon\sqrt{|n|})$. Then $|\bar\beta| = |n|/\beta \leqslant \sqrt{|n|}$. So $|u| = \frac{|\beta + \bar\beta|}{2} \leqslant \frac{(\varepsilon+1)\sqrt{|n|}}{2}$ and $|v| \leqslant \frac{(\varepsilon+1)\sqrt{|n|}}{2\sqrt{d}}$, leaving finitely many classes.

**Hint 2.** Counting: in each class the positive solutions are $\beta\varepsilon^k$ with $k$ large enough, and $x \approx \frac{1}{2}\beta\varepsilon^k$. So $x \leqslant X$ holds for $\log_\varepsilon X + O(1)$ values of $k$. Summing over the $h$ classes gives $\frac{h}{\ln \varepsilon} \ln X + O(1)$.

**Hint 3.** For $x^2 - 3y^2 = 13$ we have $\varepsilon = 2 + \sqrt{3}$. Search $\beta$ in the box from Hint 1: the classes are represented by $4 + \sqrt{3}$ and $5 + 2\sqrt{3}$. All positive solutions are $(4 + \sqrt{3})(2 + \sqrt{3})^k$ and $(5 + 2\sqrt{3})(2 + \sqrt{3})^k$, $k \geqslant 0$: $(4, 1), (5, 2), (11, 6), (16, 9), (40, 23), (59, 34), \ldots$

## Problem 9

**Hint 1.** Lagrange: if $\alpha$ is a quadratic irrational, its complete quotients have the form $\alpha_k = (P_k + \sqrt{D})/Q_k$ with $Q_k \mid D - P_k^2$. Show that $P_k, Q_k$ are bounded, so some $\alpha_k$ repeats. The converse ($\alpha$ periodic implies quadratic) is the fixed point of a Möbius map.

**Hint 2.** Galois: call $\alpha$ reduced if $\alpha > 1$ and $-1 < \bar\alpha < 0$. Show that $\alpha$ reduced implies $\alpha_1$ reduced, and that a reduced number determines its predecessor ($-1/\bar\alpha_{k+1}$ has integer part $a_k$). So the period has no pre-period. Conjugation reverses the period.

**Hint 3.** $\sqrt{d} + a_0$ is reduced, which gives $\sqrt{d} = [a_0; \overline{a_1, \ldots, a_{r-1}, 2a_0}]$; the palindrome comes from reversal. Compute: $\sqrt{m^2 + 1} = [m; \overline{2m}]$ with $\eta = m + \sqrt{m^2 + 1}$ (norm $-1$) and $\varepsilon = \eta^2$. Also $\sqrt{m^2 - 1} = [m - 1; \overline{1, 2m - 2}]$ with $\eta = \varepsilon = m + \sqrt{m^2 - 1}$.

## Problem 10

**Hint 1.** Write $a(a + 1) = Du^2$ with $D$ squarefree. Then $(a^2 + a)(b^2 + b)$ is a square if and only if $a$ and $b$ have the same $D$. Also $(2a + 1)^2 - 4Du^2 = 1$, a Pell equation.

**Hint 2.** The diagonal $a = b$ gives $N$ pairs. An off-diagonal pair needs two solutions of $X^2 - 4DU^2 = 1$ with $X \leqslant 2N + 1$. This forces the fundamental solution to be $\lesssim \sqrt{N}$, so only $O(\sqrt{N})$ values of $D$ qualify, each with $O(\log N)$ solutions.

**Hint 3.** Hence $S_N = N + O(\sqrt{N} \log^2 N)$ and $\ell = 1$. For the second question: $b = 4a(a + 1)$ pairs with $a$, since $a(a + 1)b(b + 1) = \big(2a(a + 1)(2a + 1)\big)^2$. This gives $\gg \sqrt{N}$ off-diagonal pairs, so $|S_N - N| \leqslant N^{1/2025}$ is *false*.